Cloud Computing - Class Notes

1. What is Cloud Computing?

Cloud computing means using computing services such as servers, storage, databases and software over the internet instead of owning physical hardware. Users pay only for what they use, and resources can be increased or reduced quickly.

2. Service Models

IaaS (Infrastructure as a Service) gives virtual machines and storage. PaaS (Platform as a Service) gives a ready platform to build and run apps. SaaS (Software as a Service) gives complete software through a browser, such as email or online document tools.

3. Deployment Models

A public cloud is shared and owned by a provider. A private cloud is used by one organization. A hybrid cloud combines both. A community cloud is shared by organizations with common needs.

4. Key Benefits

Lower upfront cost, easy scaling, high availability, automatic backups and access from anywhere. Teams can launch new projects in minutes instead of weeks.

5. Challenges

Data security and privacy, dependence on internet connection, vendor lock-in and unexpected costs when usage grows. Good monitoring and budgeting help control these risks.

6. Revision Points

Remember the three service models: IaaS, PaaS, SaaS. Remember the four deployment models: public, private, hybrid, community. Exam tip: give one real example for each model.